

PENDENNIS POINT, United Kingdom

WMO: 038100

Lat: 50.1452N

Lon: 5.0454W

Elev: 138

StdP: 14.62

Time Zone: 0.00 (EUW)

Period: 94-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	34.2	36.3	25.6	19.7	38.4	27.8	21.9	38.5	58.3	50.3	50.6	50.5	10.2	30	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	71.3	64.3	69.1	62.8	67.2	61.6	65.5	69.3	64.1	67.5	62.9	65.7	8.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.3	67.1	62.7	85.7	65.6	61.6	82.4	64.6	30.3	69.6	29.3	67.6	28.4	65.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 46.3	39.9	33.3	DB	31.5	76.3	1.4	3.5	30.5	78.8	29.7	80.8	29.0	82.7	28.0	85.3	(4)
(5)			WB	29.5	68.3	1.3	2.6	28.6	70.2	27.9	71.7	27.2	73.2	26.3	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	46.4	45.4	46.7	49.6	53.3	58.0	61.5	62.6	59.6	55.4	50.5	47.0	(6)	
	DBStd	6.93	4.00	4.14	3.76	3.36	3.13	2.94	3.03	2.86	2.74	3.10	3.57	4.31	(7)	
	HDD50	565	120	132	109	47	8	0	0	0	0	3	37	109	(8)	
	HDD65	4381	578	549	566	463	361	210	114	88	162	298	434	558	(9)	
	CDD50	1681	7	3	8	34	112	241	358	390	289	170	52	17	(10)	
	CDD65	22	0	0	0	0	0	1	7	13	1	0	0	0	(11)	
	CDH74	23	0	0	0	0	0	0	4	19	0	0	0	0	(12)	
	CDH80	2	0	0	0	0	0	0	0	2	0	0	0	0	(13)	
(14) Wind	WSAvg	14.0	17.8	15.7	15.4	13.2	13.2	11.1	10.3	10.3	12.7	16.3	15.0	17.2	(14)	
(15) Precipitation	PrecAvg	44.0	5.3	4.0	3.7	2.6	2.4	2.4	2.4	2.9	3.5	4.4	5.0	5.3	(15)	
	PrecMax	58.0	11.2	8.3	7.2	5.8	6.3	6.1	5.2	6.1	6.8	8.4	9.1	10.5	(16)	
	PrecMin	34.8	0.7	0.2	0.1	0.6	0.2	0.2	0.5	0.2	1.1	0.6	2.2	1.4	(17)	
	PrecStd	5.7	2.4	2.2	1.7	1.3	1.3	1.6	1.2	1.5	1.6	2.1	2.0	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	54.0	57.1	60.4	66.0	70.1	74.0	76.0	71.0	64.1	59.1	55.8	(19)	
		MCWB	51.8	51.3	51.9	53.7	59.7	61.5	65.1	68.3	63.8	58.4	56.0	54.4	(20)	
	2%	DB	53.5	52.9	54.0	58.2	62.6	67.3	71.1	71.7	67.2	61.6	57.4	54.8	(21)	
		MCWB	51.7	50.6	50.1	52.4	56.4	60.5	63.9	65.5	61.5	57.8	55.2	53.5	(22)	
	5%	DB	52.6	52.0	52.5	56.4	60.2	65.3	69.1	69.2	65.4	60.3	56.5	54.0	(23)	
		MCWB	51.0	50.1	49.7	51.3	55.1	59.2	62.9	63.4	60.6	57.3	54.5	52.5	(24)	
	10%	DB	51.7	51.0	51.5	54.7	58.4	63.5	67.0	67.2	64.1	59.4	55.6	53.0	(25)	
		MCWB	50.3	49.2	49.2	50.6	53.9	58.2	61.7	61.8	60.2	57.0	53.7	51.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.9	52.7	55.6	60.0	62.6	67.0	69.6	65.2	60.6	58.0	55.0	(27)	
		MCDB	53.5	53.4	54.6	58.7	65.2	68.1	70.9	75.4	68.3	62.1	58.3	55.2	(28)	
	2%	WB	52.2	51.8	51.7	53.9	57.6	61.2	64.8	66.4	63.4	59.3	56.3	54.1	(29)	
		MCDB	52.8	52.4	52.8	56.3	61.0	65.7	68.9	70.6	65.5	60.2	56.8	54.5	(30)	
	5%	WB	51.3	50.8	50.8	52.6	56.1	60.1	63.7	64.7	62.1	58.6	55.2	53.1	(31)	
		MCDB	52.1	51.6	51.9	55.0	58.9	63.9	67.6	68.1	64.0	59.5	56.0	53.7	(32)	
	10%	WB	50.4	49.6	49.9	51.4	54.9	59.1	62.5	63.3	61.1	57.9	54.0	51.9	(33)	
		MCDB	51.4	50.7	51.0	53.8	57.3	62.5	65.8	66.0	63.1	59.0	55.2	52.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.8	6.7	8.3	8.4	8.8	8.6	8.9	8.5	6.7	6.2	6.1	(35)	
		MCDBR	5.8	6.9	7.8	10.7	10.8	11.5	11.8	11.8	10.5	7.8	6.6	6.1	(36)	
	5% WB	MCWBR	5.8	6.6	5.7	7.0	6.8	7.0	6.4	6.6	6.2	5.8	6.0	6.5	(37)	
		MCWBR	5.6	6.1	6.8	9.2	9.2	10.4	10.1	10.2	8.4	6.4	5.6	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.339	0.352	0.380	0.386	0.393	0.380	0.384	0.369	0.366	0.360	0.337	0.331	(40)	
		taud	2.354	2.375	2.316	2.329	2.352	2.420	2.416	2.475	2.468	2.443	2.404	2.320	(41)	
	Edn at Noon	Edn at Noon	217	247	260	272	274	279	275	275	263	241	217	199	(42)	
		Edh at Noon	22	27	33	36	37	35	35	31	29	25	20	19	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	263	514	880	1425	1721	1886	1735	1489	1131	645	338	212	(44)	
	RadStd	RadStd	23	51	96	95	125	175	116	104	76	41	26	23	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page